Multimodal data and fusion

Multimodal data

Medical image



pixels

Radiology report

FINDINGS
Lungs are clear. Heart size is normal. No effusion or pneumothorax.

IMPRESSION
No acute cardiopulmonary disease.

free text

Labs and demographics

Age	54
Sex	F
WBC	11.2 $\times 10^9$/L
CRP	24 mg/L
Hb	13.1 g/dL

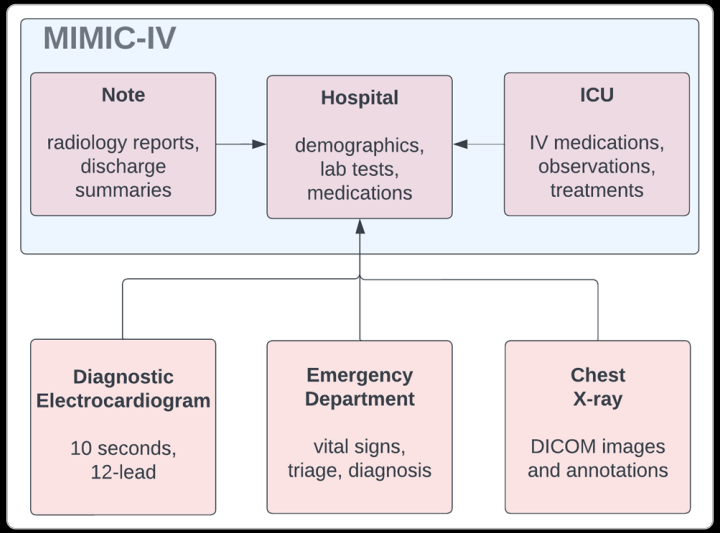
numbers and categories

one patient

Biobanks, health records, imaging and wearable sensors now set the stage for **multimodal AI** that captures the **complexity of human health and disease**.

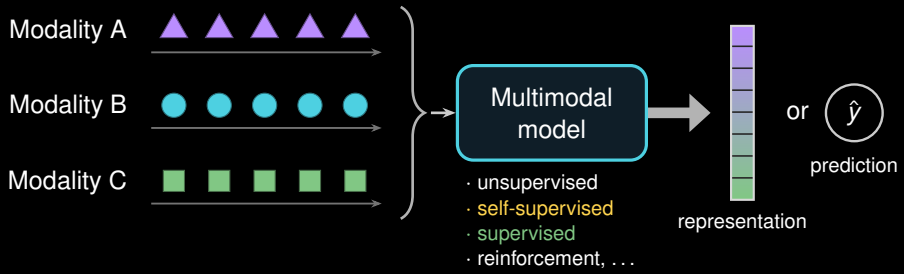
Acosta et al., *Multimodal biomedical AI*, Nature Medicine

MIMIC-IV, A FREELY ACCESSIBLE ELECTRONIC HEALTH RECORD DATASET



Johnson et al., *MIMIC-IV, a freely accessible electronic health record dataset*, Scientific Data

MULTIMODAL CLINICAL AI

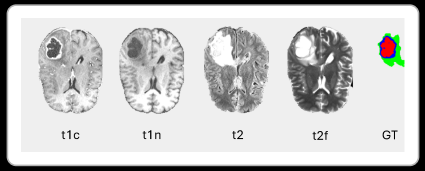


Morency & Liang, *Multimodal Machine Learning*, CMU 11-777

More on multimodal data

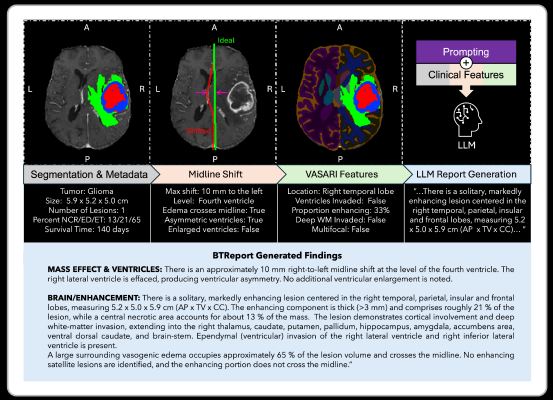
Special case

Nearly identical MRIs from tumor segmentation tasks.



Common case

MRI and radiology report.

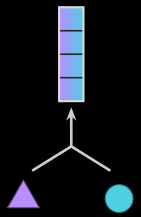


Ren et al., *Here Comes the Explanation: A Shapley Perspective on Multi-contrast Medical Image Segmentation*, arXiv; Heras Rivera et al., *BTReport: A Framework for Brain Tumor Radiology Report Generation with Clinically Relevant Features*, MIDL

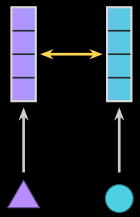
THREE DIFFERENT STRATEGIES IN MULTIMODAL LEARNING

Representation: learning representations that reflect cross-modal interactions between individual elements, across different modalities.

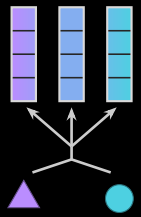
Fusion



Coordination

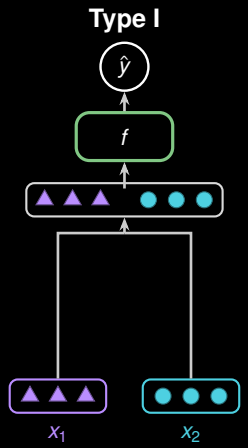


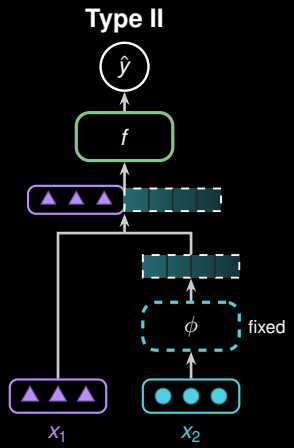
Fission



Early, joint and late fusion

EARLY FUSION





$\hat{y} = f([x_1 ; x_2])$ $\min_f \mathcal{L}$

Original inputs, concatenated.

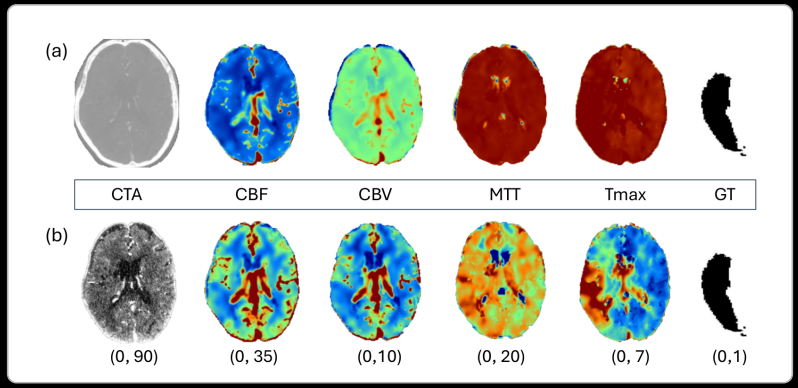
$\hat{y} = f([x_1 ; \phi(x_2)])$ $\min_f \mathcal{L}$

One input pre-extracted by a fixed ϕ.

Fuse before the model; the loss trains only f.

Huang et al., *Fusion of medical imaging and electronic health records using deep learning*, npj Digital Medicine

Early fusion: stack the modalities as channels



CTA CT angiogram: the blood vessels

CBF cerebral blood flow: how fast blood reaches the tissue

CBV cerebral blood volume: how much blood is in the tissue

MTT mean transit time: how long blood takes to pass through

Tmax time to maximum: how late the contrast arrives

GT ground truth: the lesion, traced on MRI days later

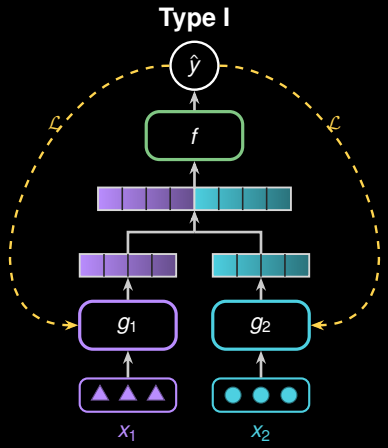
Row (a) is each map as acquired; row (b) is the same map after windowing.

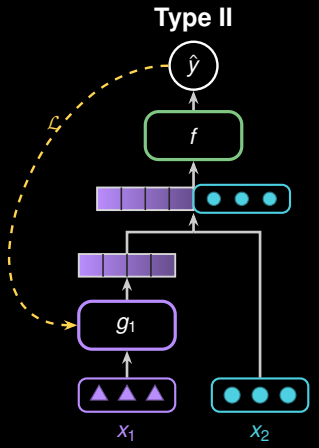
Heras Rivera et al., *Clinically-Informed Preprocessing Improves Stroke Segmentation in Low-Resource Settings*, MICCAI MIRASOL Workshop

Limitations of input-level stacking

- **Incompatible spatial data.** A 3D MRI volume and a 2D X-ray projection differ in dimensionality and spatial correspondence, preventing direct channel-wise stacking.
- **Heterogeneous data types.** Clinical notes, laboratory measurements, and medication records lack the spatial structure required for direct stacking with image inputs.
- **Missing modalities at inference.** Models trained on a fixed set of input channels generally require all corresponding modalities at inference.
- **Alternative: joint fusion.** Modality-specific encoders map each input to a feature representation, which can then be combined for downstream prediction.

JOINT FUSION





$$\hat{y} = f([\,g_1(x_1)\,;\,g_2(x_2)\,]) \quad \min_{f,\,g_1,\,g_2} \mathcal{L}$$

Every modality gets a trained encoder.

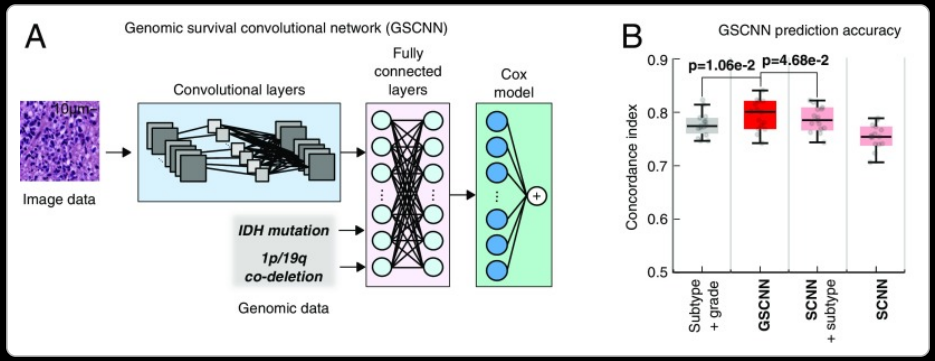
$$\hat{y} = f([\,g_1(x_1)\,;\,x_2\,]) \quad \min_{f,\,g_1} \mathcal{L}$$

One encoder; the other input joins raw.

Fuse learned **features**; the loss reaches every encoder.

Huang et al., *Fusion of medical imaging and electronic health records using deep learning*, npj Digital Medicine

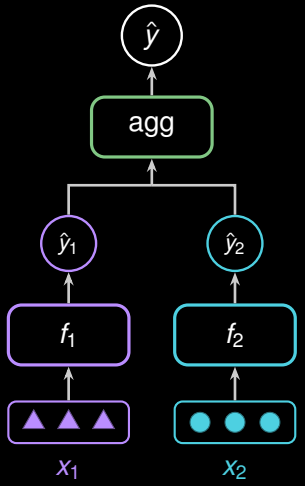
JOINT FUSION: HISTOLOGY AND GENOMICS



For glioma survival, IDH and 1p/19q status enter the CNN's fully connected layers and one Cox loss trains the whole network (**joint fusion, Type II**), raising the median c-index from 0.754 (histology alone) to **0.801**.

Mobadersany et al., *Predicting cancer outcomes from histology and genomics using convolutional networks*, PNAS

Late fusion



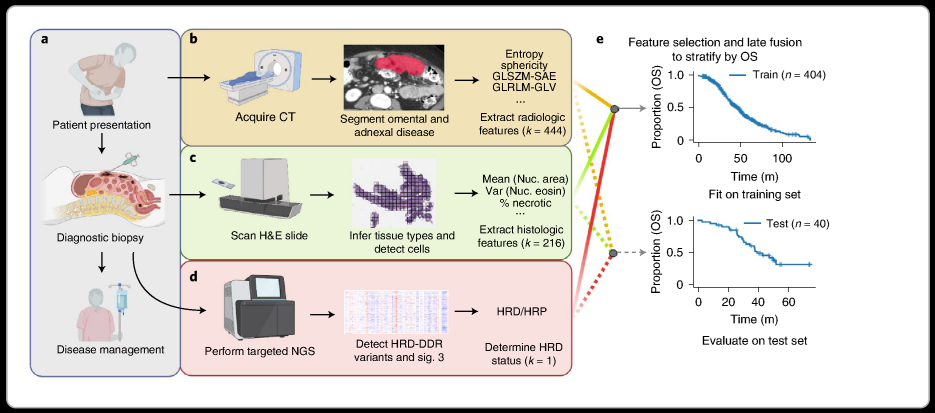
$$\hat{y} = \mathrm{agg}\big(f_1(x_1), f_2(x_2)\big)$$

$$\min_{f_1} \mathcal{L}(\hat{y}_1, y) \qquad \min_{f_2} \mathcal{L}(\hat{y}_2, y)$$

Fuse **predictions**. Each modality is its own model, trained alone; agg is a fixed rule such as an average, or a small trained model.

Huang et al., *Fusion of medical imaging and electronic health records using deep learning*, npj Digital Medicine

LATE FUSION: CT, HISTOLOGY AND GENOMICS

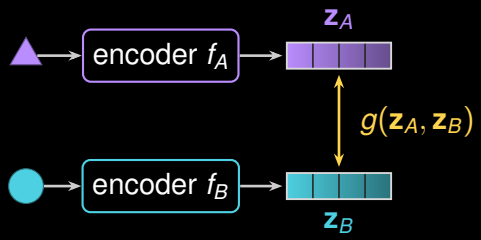


For ovarian cancer survival, each modality gets its own Cox model and a final Cox model fuses their risk scores. The best test model used **CT and histology alone** (c-index 0.62); adding all four modalities did **worse**. More modalities are no guarantee.

Boehm et al., *Multimodal data integration using machine learning improves risk stratification of high-grade serous ovarian cancer*, Nature Cancer

Coordination

Coordination: two encoders, one similarity



Learning with a coordination function:

$$\mathcal{L} = g\big(f_A(\blacktriangle),\ f_B(\bullet)\big)$$

Trained jointly over θ_g, θ_{f_A} and θ_{f_B}.

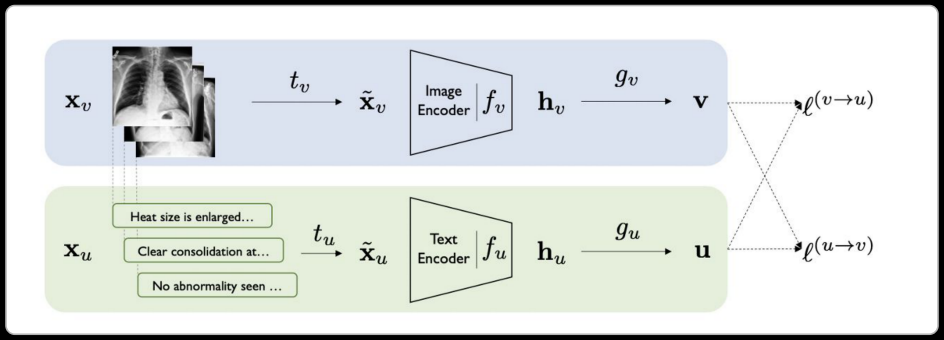
Requires paired data: every triangle needs its own circle.

Example: cosine similarity

$$g(\mathbf{z}_A, \mathbf{z}_B) = \frac{\mathbf{z}_A \cdot \mathbf{z}_B}{\|\mathbf{z}_A\|\,\|\mathbf{z}_B\|}$$

1 when the two embeddings point the same way

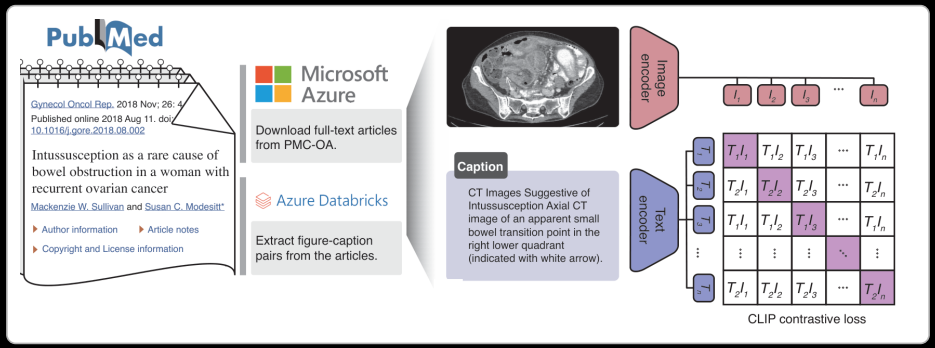
COORDINATION: CONVIRT, X-RAYS AND REPORTS



217k image and text pairs from MIMIC-CXR. A **ResNet-50** encodes the image, a **BERT** text encoder initialized from ClinicalBERT encodes the report, each with its own projection head. The loss runs in both directions, image to text and text to image, so neither modality is merely a target.

Zhang et al., *Contrastive Learning of Medical Visual Representations from Paired Images and Text*, MLHC

COORDINATION: BIOMEDCLIP AT BIOMEDICAL SCALE



15 million figure and caption pairs from 4.4 million PubMed Central articles. An image encoder and a text encoder are trained together: each image is pulled towards **its own caption** and pushed away from every other caption in the batch. It beats the radiology-specific BioViL on RSNA pneumonia detection.

Zhang et al., *BiomedCLIP: a multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs*, arXiv

Self-supervised learning

Challenges in medical data annotation

- Complexity and ambiguity in medical imaging: findings can be subtle and open to interpretation, so consistent annotations are hard.
- Shortage of qualified annotators with medical expertise: labels need certified radiologists, pathologists or clinical experts.
- Time-intensive and costly annotation processes: a single CT or MRI scan can take hours, especially for segmentation.

Voronovska, *Top 5 Challenges in Medical Data Annotation*, medDARE

Learning without labels

- Every label needs an expert. A segmentation mask can take a doctor an hour for one scan.
- So labelled medical datasets stay small: a thousand images is a big study. ImageNet has 1.2 million.
- But hospitals already store hundreds of thousands of scans that nobody has labelled.
- The idea: make up a task the scan can **check by itself**. Then the whole archive becomes training data, with no one labelling anything.
- It is catching on fast: medical imaging papers using this idea grew more than tenfold from 2019 to 2021.

Huang et al., *Self-supervised learning for medical image classification: a systematic review and implementation guidelines*, npj Digital Medicine

What is Self-Supervised learning?

- Supervised learning: train the model directly on the task you care about, using ground truth labels: class labels, expert masks, outcomes from the record.
- Self-supervised learning: do *not* train on the task directly.
  - Invent a **pretext task** whose answer is generated from the raw data itself.
  - Use it to learn a good feature encoder. No external labels are used at this stage.
  - Then fine-tune that encoder on the real task, with far fewer labels.

What Are the Advantages of Self-Supervised Learning?

- Scarce labels: where labelling needs an expert, the label is the bottleneck. Self-supervision puts to work the data that would otherwise sit unused for want of a label.
- Data at scale: training scales with the archive, not with how much annotation you can buy.
- Versatile: the same recipe learns from text, speech and images. What comes out is a general encoder, not a solution to one task.
- Robust and general: trained on broader data, the model overfits less. That matters when deployment is diverse and unpredictable.